# Advanced Deep Learning — Project Proposals

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## Proposal: Flow matching for image generation

### Goal and modelization

The goal of this project is to implement and study *Conditional Flow Matching (CFM)*, a recent generative modeling framework that learns a continuous vector field $v\theta(x, t)$ transporting samples from a Gaussian noise distribution (t=0) to the data distribution (t=1). Unlike diffusion models, which follow curved stochastic paths during generation, flow matching with optimal transport coupling defines straight-line interpolations $x_t = (1-t)z_0 + tx_1$ between noise $z_0$ and data $x_1$, and trains a neural network to regress the velocity $u = x_1 - z_0$ at each point along the path. At inference, new samples are generated by solving the ODE $dx/dt = v\theta(x, t)$ from t=0 to t=1 using a standard ODE solver (Euler or Runge-Kutta). The straight-path formulation means fewer integration steps are needed compared to diffusion models, making generation significantly faster. We will implement a U-Net-based flow matching model and systematically compare it against DDPM and DDIM on image generation quality (FID) as a function of the number of sampling steps.

### Link with Course Concepts

This project connects to several course chapters. **Session 7 (Generative Models):** flow matching is a direct successor to diffusion models and normalizing flows, both covered in the course; we compare all three frameworks. **Session 1 (Optimization):** the training involves a regression loss on the velocity field, and generation requires choosing an ODE solver and step size — a tradeoff between speed and accuracy that we study systematically. **Session 4 (Frugal AI):** the core advantage of flow matching is computational efficiency at inference — achieving comparable quality with 5–10× fewer neural network evaluations.

### Dataset

We will use **CIFAR-10** (60,000 32×32 color images, 10 classes) as the primary benchmark, which is the standard dataset for comparing generative models. For 2D toy visualizations of the learned vector field, we will use synthetic datasets (moons, circles, spirals) generated with scikit-learn.

### Experiment Plan and Baselines

We will train and compare: (1) a **DDPM** baseline with 1000 denoising steps, (2) **DDIM** sampling at 10, 20, 50, 100 steps (accelerated diffusion sampling without retraining), and (3) our **Conditional Flow Matching** model with Euler and RK4 solvers at 1, 2, 4, 8, 16, 50 ODE steps. All models will use the same U-Net backbone for fair comparison. The main evaluation metric is FID score (Fréchet Inception Distance) as a function of the number of function evaluations (NFE). We will additionally visualize: (a) the learned 2D vector fields on toy data, (b) generation trajectories showing noise-to-image evolution, (c) latent space interpolations, and (d) conditional generation using class labels.

### References

[1] Lipman, Y., Chen, R.T.Q., Ben-Hamu, H., Nickel, M. *Flow Matching for Generative Modeling*. ICLR, 2023.

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[3] Ho, J., Jain, A., Abbeel, P. *Denoising Diffusion Probabilistic Models*. NeurIPS, 2020.

[4] Song, J., Meng, C., Ermon, S. *Denoising Diffusion Implicit Models*. ICLR, 2021.

[5] Tong, A., Malkin, N., Huguet, G., Zhang, Y., et al. *Improving and Generalizing Flow-Based Generative Models with Minibatch Optimal Transport*. TMLR, 2024.